CODY KICKERTZ

AI Systems Architect · Platform & Infrastructure

Rust · agent runtimes · governed AI infrastructure · distributed systems

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SELECTED SYSTEMS · PUBLIC REPOSITORIES UNLESS MARKED PRIVATE

Aletheia — Self-hosted multi-agent AI runtime with embedded Datalog and HNSW state over a knowledge graph; eleven-factor recall; per-tool Landlock, seccomp, and network-namespace controls; multi-provider LLM layer with circuit breaking and loop detection. Ships as one binary with embedded state; inference uses a configured LLM provider, and optional tools or messaging channels can open network paths.

Kanon — **Private case study.** Standards-and-dispatch control plane for a multi-repository fleet: standards-as-code linting, Rust-analyzer HIR checks, MCP tooling, git forge, CI, and a compare-and-swap work queue. Public receipt: six featured public repositories carry repository-owned .kanon-ci.toml; enforcement scope differs by repository.

Logismos — GPU inference stack for AMD RDNA3 in progress: hand-written HIP FFI and WMMA kernels, a custom GGUF loader, and attention from first principles. Verified end to end on CPU; GPU cutover remains scoped and blocked on hardware access.

Hamma — Clean-room Rust implementation of a Tailscale-compatible control protocol—not a translation of the Go source. Noise IK handshake, control-plane wire types, TCP/TLS registration, and map streaming have landed with byte-level tests; the BoringTun-backed WireGuard data plane is planned.

Thumos — From-scratch bare-metal OS for an Armv7 smartphone target. Own MMU, log-structured filesystem, Ed25519 measured boot, and Signal protocol implementation; treats the cellular baseband as an untrusted peripheral behind a driver boundary. Cross-compiled and QEMU-tested.

Harmonia — Self-hosted media platform: custom QUIC renderer protocol with NTP-style clock sync, lock-free audio ring buffer, biquad DSP pipeline, and a headless Raspberry Pi renderer through a NixOS AArch64 cross-build.

EXPERIENCE

Summus Global · Data Scientist & AI Systems Architect — 2023–Present

- Architected an agent-native internal analytics platform in Rust: a governed MCP tool server with 137 tools over the data warehouse, protected by SQL validation, SELECT-only database policy, and a hard block on raw-CLI writes against production PHI.
- Built a standards-as-code governance layer with 284 machine-enforced rules across 13 registries, wired to commit-time gates and a promotion loop that turns observed tool use into reviewed, durable infrastructure.
- Designed a medical-code taxonomy engine spanning 301,000 ICD-10, CPT, HCPCS, and SNOMED codes in a four-level, 7,771-node hierarchy, projected in 12 seconds; implemented a multi-task LoRA fine-tune and verified its LoRA-to-Candle conversion without application-code changes.
- Replaced a 40-cell Hex/Python ROI-scoring notebook with a production Rust engine at exact parity on a $4.18M validation client; identified three compounding legacy scoring defects and rebased a flagship analysis onto a third-party-validated basis.
- Added a claims-manifest gate pairing each client-facing figure with a runnable, tolerance-bound SQL check before a report can ship across 30+ ROI engagements.

Summus Global · Database, Data & Business Intelligence Analyst — 2023–2025

- Designed an AWS Redshift dimensional model using star/snowflake patterns, CTEs, and window functions, powering eight dashboards and 40+ ROI analyses across the business.
- Built a React/FastAPI medical-taxonomy system with 50+ endpoints and PostgreSQL, Redis, Qdrant, and Elasticsearch, serving 192,000 hierarchical records with semantic search.
- Built physician deduplication with fuzzy matching and NPI reconciliation, raising NPI coverage from 33% to 92% across 1,200+ profiles and unblocking three dependent product features.

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LEADERSHIP & EARLIER EXPERIENCE

United States Marine Corps — Captain · Camp Lejeune, NC 2018–2023

Assistant Disbursing Officer · Dec 2022–May 2023

- Helped lead a disbursing office of 157 Marines and 12 civilians across seven finance functions.
- Reengineered core disbursing processes: 16% faster processing, 14% accuracy gain, roughly 375 hours returned annually, and a 3,000+ claim backlog eliminated across $45M+ in military pay and travel claims.

Disbursing Officer, 22d Marine Expeditionary Unit · Aug 2021–Dec 2022

- Ran fiscal operations for a 3,000-person MEU through a seven-month deployment aboard three naval vessels.
- Managed a $350,000 deployed cash budget with zero discrepancies; consolidated three accounting systems into one, later adopted by East Coast MEU deployments.

Travel Section Officer-in-Charge · Dec 2018–Aug 2021

- Led a 62-person travel-claims section supporting service members, civilians, and dependents across the eastern United States and Europe.
- Reduced claim backlogs 30% through workload triage, process redesign, and production tracking.

Clarkson Aerospace Corp · Cyber Security Research Intern · Houston, TX 2015–2018

- Built a predictive intrusion-detection system for the Air Force Research Laboratory using PCA, k-medoids clustering, and SVM classification in R; reached 98% accuracy on a 125,000-observation network-traffic dataset.
- Selected to present the research at Pentagon Lab Day and the Georgia Institute of Technology.

TECHNICAL

Languages	Rust (primary), Python, SQL, TypeScript, R, Bash/Fish, Nix, Typst
Agent / AI	MCP tool servers, agent orchestration and dispatch, standards-as-code governance, RAG, embeddings with Candle and MedEmbed, LoRA/MNRL/Matryoshka fine-tuning, clinical NLP
Systems	Tokio async, fjall/LSM, QUIC/Quinn, Landlock/seccomp, WireGuard/Noise, no_std/embedded, HIP/GPU kernels
Data	AWS Redshift, Polars, SQLite, dimensional modeling, ETL pipelines, Hex in HIPAA environments
Infrastructure	Podman, systemd, Tailscale, NixOS, GitHub Actions, restic, nftables

EDUCATION

The University of Texas at Austin — McCombs School of Business · MBA May 2026

University of Houston — College of Technology · BS, Computer Information Systems